

Friday 18/12/2020

Midterm

Duration: 90 minutes

P1 [20 points] Construct a DFA for the languages

- (a) Binary strings that contain neither 00 nor 11.
- (b) $L = \{w \mid w \in \{0,1\}^* \text{ where all 1s are in groups of 2 or 4.}\}$ For example $001100111101100 \in L$ but $001110011 \notin L$ because some 1s are in a group of 3.

P2 [20 points] Construct an NFA for the languages.

- (a) Binary strings where every third position (3rd, 6th, 9th, etc. positions) contains a 1. (Use three states, every extra state halves your points)
- (b) The language $1^*(001^+)^*$. (Use three states, every extra state halves your points)

P3 [25 points] For all the languages below, decide whether they are regular or not and prove your answer.

- (a) $L = \{a^k b^m c^n \mid \text{where } k + m + n = 100\}$
- (b) $L = \{0^m 1^n \mid m, n \geq 0, m \neq n\}$
- (c) $L = \{1^k y \mid y \in \{0,1\}^* \text{ and } y \text{ contains at most } k \text{ 1s, for } k \geq 1\}$

P4 [25 points] For each of the languages below, decide whether they are context-free or not. If yes, give a context-free grammar and a pushdown automata for the following languages. Otherwise prove that it is not context-free. (Do not forget that PDAs can be non-deterministic.)

- (a) Binary strings of odd length where the first, middle and last symbols are the same.
- (b) $\{a^m b^n \mid m \leq n \leq 3m\}$.
- (c) $\{a^m b^n \mid m, n \geq 1 \text{ and } m \text{ divides } n\}$

P5 [10 points] Consider a TM variant HopTM where the head can only go 3 cells forward (right) or 2 cells back (left). Prove that HopTM is equally powerful as an ordinary TM.